

Candidate's seat No: _____

Charotar University of Science and Technology [CHARUSAT]
Faculty of Technology and Engineering
Department of Mathematical Sciences
MA243.01 TRANSFORMS AND VECTOR DIFFERENTIAL CALCULUS
2nd UNIT TEST

Semester: 3rd Sem. B. Tech. (EC)

Maximum Marks: 30

Date : 08/10/2018 (Monday)

Time: 09:15 am to 10:15 am

Instructions:

- (i) Figures to the right indicate **full** marks.
- (ii) Use of scientific calculator is allowed.
- (iii) Draw the figure where it is required.

Q-1 Choose the correct answer from the given options in the followings: [06]

1. The convolution of 1 and 1 is _____.
(a) 1 (b) t
(c) t^2 (d) None of these
2. The value of $\int_0^{2\pi} \cos^2 3x \, dx =$ _____.
(a) 0 (b) π
(c) $-\pi$ (d) None of these
3. Which of the following is not Dirichlet's condition?
(a) $f(x)$ is periodic. (b) $f(x)$ is finite.
(c) $f(x)$ is single valued function. (d) None of these
4. The value of $Z\left(\cos \frac{n\pi}{2}\right)$ is _____.
(a) $\frac{z^2}{z^2+1}$ (b) $\frac{z}{z^2+1}$
(c) $\frac{-z}{z^2+1}$ (d) None of these
5. The value of $Z\left(\left(\frac{1}{2}\right)^n\right)$ is _____.
(a) $\frac{2z}{2z+1}$ (b) $\frac{z}{2z+1}$
(c) $\frac{-z}{z^2+1}$ (d) None of these
6. The inverse Fourier transform of $f(x)$ is defined as _____.
(a) $\frac{1}{2\pi} \int_{-\infty}^{\infty} F(s)e^{isx} ds$ (b) $\frac{1}{2\pi} \int_0^{\infty} F(s)e^{isx} ds$
(c) $\frac{1}{2\pi} \int_{-\infty}^{\infty} F(s)e^{-isx} ds$ (d) $\frac{1}{2\pi} \int_0^{\infty} F(s).e^{-isx} ds$

Q-2 Attempt any Three.**[12]**

- (a) I. Find the Z transform of $Z(n^2 a^n)$.
II. Find the Z transform of $Z\left(\frac{1}{n}\right)$.
- (b) I. Find the Fourier series of $f(x) = x$ in the interval $-\pi < x < \pi$.
II. Find the Fourier transform of $f(x) = e^{-a^2 x^2}$, $a < 0$.
- (c) Solve $y'' + 3y' + 2y = e^t$ with $y(0) = 1$ and $y'(0) = 0$.
- (d) Using convolution theorem, find the inverse Laplace Transform of $\frac{s}{(s^2 + 4)^2}$.

Q-3 Attempt any Three.**[12]**

- (a) Obtain the Fourier series to represent $f(x) = x^2$ in the interval $-\pi < x < \pi$. Deduce $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$ from it.
- (b) Find the Fourier series for
$$f(x) = \begin{cases} x & ; 0 \leq x \leq \pi \\ 2\pi - x & ; \pi \leq x \leq 2\pi \end{cases}$$
- (c) Define Fourier sine transform. Find the Fourier transform of
$$f(x) = \begin{cases} 1 - x^2, & |x| \leq 1 \\ 0 & , |x| > 1 \end{cases}$$
- (d) State convolution theorem for Fourier transforms. Find the Fourier cosine transform of $f(x) = 5e^{-3x} + 2e^{-2x}$.
